

```

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
print("Upload an image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Add Artificial Gaussian Noise
# -----

noise = np.random.normal(0, 25, img.shape).astype(np.int16)

noisy = img.astype(np.int16) + noise
noisy = np.clip(noisy, 0, 255).astype(np.uint8)

# -----
# Remove Noise using Gaussian Blur
# -----

denoised = cv2.GaussianBlur(noisy, (5,5), 0)

# -----
# Display Images
# -----

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(noisy)
plt.title("Noisy Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(denoised)
plt.title("Denoised Image")
plt.axis("off")

plt.show()

# -----
# Explanation
# -----

print("Image Acquisition Noise Sources:")
print("- Low light")
print("- Sensor electronic noise")
print("- High ISO")
print("- Thermal noise")
print("- Transmission errors")

print("\nNoise Reduction Techniques:")

```

```
print("- Gaussian Filter")
print("- Median Filter")
print("- Better camera sensors")
print("- Improved lighting")
print("- Lower ISO settings")
```

Upload an image

WIN_20260..._48_Pro.jpg

WIN_20260609_09_49_48_Pro.jpg(image/jpeg) - 225783 bytes, last modified: 09/06/2026 - 100% done

Saving WIN_20260609_09_49_48_Pro.jpg to WIN_20260609_09_49_48_Pro.jpg

Original Image



Noisy Image



Denoised Image



Image Acquisition Noise Sources:

- Low light
- Sensor electronic noise
- High ISO
- Thermal noise
- Transmission errors

Noise Reduction Techniques:

- Gaussian Filter
- Median Filter
- Better camera sensors
- Improved lighting
- Lower ISO settings